Mariely Ann L. Tolentino John Randolf M. Penaredondo, MIT

Week 2 Portfolio Project

PART 1: Company Profile

1. Company Name
- ABC Startup Solutions

2. Nature of Business
- ABC Startup Solutions is a software development company that provides:
    - Custom software development
    - Web application development
    - Mobile application development
    - Cloud-based application deployment
    - Software testing and quality assurance
    - Basic IT support and technology consulting for small businesses

- The company primarily serves small and medium enterprises in the Laguna and CALABARZON area.

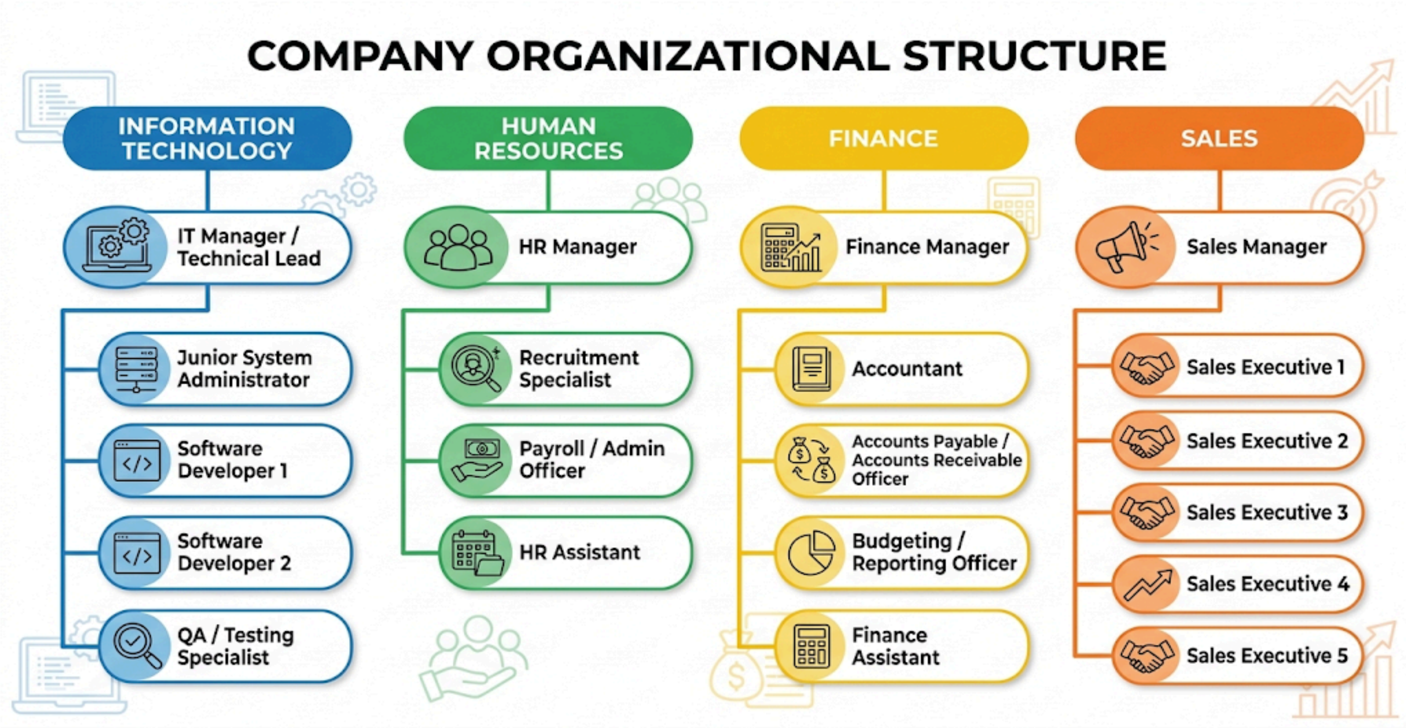
3. Company Vision
- To become a trusted and innovative software development partner that delivers secure, reliable, and scalable technology solutions for growing businesses in the Philippines.

4. Company Mission
- Our mission is to:
    - Develop high-quality software solutions that meet client needs.
    - Provide a secure, efficient, and modern IT environment for our employees.
    - Support business growth through reliable technology infrastructure.
    - Maintain strong data protection, system availability, and customer satisfaction.

5. Office Location
- Fictional Office Address:
    - 3rd Floor, Building B, Nuvali Tech Hub
    - Ayala Nuvali, Santa Rosa, Laguna, Philippines

- The office occupies one floor and is designed for 20 employees.

6. **Organizational Structure**



7. **Employee Distribution**

DEPARTMENT	EMPLOYEE
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

PART 2: Enterprise Hardware Inventory

The following hardware inventory is designed for a small startup office with 20 employees. The quantities are based on one primary device per employee, plus one loaner laptop for emergency use, travel, or repair. The network hardware is sized to support current operations and allow reasonable future growth.

ASSET ID	HARDWARE	QUANTITY	DEPARTMENT	PURPOSE
HW-001	Desktop Computers	14	Finance, HR, and IT	Used for daily office tasks such as document processing, spreadsheets, payroll, accounting, email, and internal file access.
HW-002	Laptops	7	Sales and IT	Used by sales employees for client meetings, presentations, proposals, customer follow-ups, remote work, and cloud-based applications.
HW-003	Server	1	IT	Hosts company server services such as file storage support, backups, DNS, DHCP, authentication, monitoring, and internal tools.
HW-004	Router	1	IT	Provides internet connection, firewall protection, VPN support, network routing, and basic network security.
HW-005	Switch	1	IT	Connects all wired devices and provides Power over Ethernet for wireless access points and future network devices.
HW-006	Printer	1	All Departments	Provides printing, scanning, and copying services for all employees in the office.
HW-007	UPS	2	IT and Finance	Provides backup power for the server, router, switch, and NAS during power interruptions.
HW-008	Wireless Access Point	2	IT	Provides reliable Wi-Fi coverage for laptops, mobile devices, and guest users across the office floor.
HW-009	NAS Storage	1	IT	Provides centralized file storage, departmental shared folders, backup storage, and file recovery

				support.
HW-010	External Backup Drive	2	IT	Used for offline backup rotation and disaster recovery protection.
HW-011	Monitors	40	All departments	Provides dual-monitor setup for each employee to improve productivity and multitasking.

Justification of Hardware Selections

The proposed hardware inventory consists of 14 desktop computers, 7 laptops, 1 server, 1 router/firewall, 1 managed switch, 1 multifunction printer, 2 UPS units, 2 wireless access points, 1 NAS storage device, 2 external backup drives, and 40 monitors. The 14 desktop computers will be allocated to Finance, Human Resources, and Information Technology because desktops are stable, secure, cost-effective, easier to maintain, and suitable for office-based tasks such as accounting, payroll, administrative work, software development, testing, virtualization, and server administration. The 7 laptops include 6 units for the Sales Department, since sales employees need portable devices for client meetings, presentations, customer visits, and remote work, and 1 loaner or travel unit that can be used when a primary device is under repair or when temporary equipment is needed. The server will provide centralized services such as file and storage coordination, DNS, DHCP, backup management, user authentication support, internal development tools, and system monitoring. The router/firewall will connect the office to the internet while protecting the network through firewall features, VPN support, and network segmentation. The 48-port Gigabit PoE+ managed switch will provide enough ports for current devices and future expansion, while also supplying power to wireless access points and other compatible network devices.

One multifunction network printer is sufficient for a single-floor office with 20 employees because it reduces cost and provides shared printing, scanning, and copying services. The two UPS units will protect critical equipment such as the server, router, switch, and NAS from sudden power loss, brownouts, and voltage fluctuations. The two wireless access points will provide better Wi-Fi coverage, stronger signal, and improved capacity for laptops, mobile phones, and other wireless devices. The NAS storage device will serve as the centralized location for department files, shared documents, software installers, backups, and project files, while the two external backup drives will support backup rotation and help protect important company data. Finally, the 40 monitors will allow each employee to use a dual-monitor setup, which improves productivity, multitasking, and efficiency, especially for IT, Finance, and Sales personnel.

PART 3: Enterprise Software Inventory

The following software inventory includes the minimum required software for the company's daily operations, software development work, system administration, security, remote support, and file management.

SOFTWARE	VERSION	LICENSE	PURPOSE
Windows 11 Pro	25H2 or latest stable build	OEM / Device license for company desktops and laptops	Primary operating system for employee computers; provides domain support, BitLocker encryption, security features, and compatibility with business software
Ubuntu Server	24.04 LTS	Open-source / Free	Server operating system for file services, backups, DNS, DHCP, authentication support, monitoring, and internal tools
Microsoft Office	Microsoft 365 Apps for Business, Current Channel	Subscription, per users	Productivity suite for documents, spreadsheets, presentations, email, and cloud collaboration
VS Code	Latest stable version	Free / Microsoft license	Code editor for software development, scripting, configuration files, and documentation
Git	Latest stable version	Open-source	Distributed version control system for tracking code changes and supporting collaborative development
GitHub Desktop	Latest stable version	Free	Graphical Git client for easier repository management and simplified version control tasks
VirtualBox	Latest stable version	Open-source / GPL	Virtualization software for testing operating systems, applications, and network configurations in a safe environment
Google Chrome	Latest stable version	Free / Chrome Enterprise	Web browser for cloud applications, online tools, research, and web application testing
Microsoft Defender	Latest platform and signature updates	Included with Windows 11 Pro	Endpoint protection against viruses, malware, ransomware, and other security threats
AnyDesk	Latest stable version	Commercial remote-support license	Secure remote access and technical support tool for

			troubleshooting user computers
7-Zip	Latest stable version	Open-source / LGPL	File compression and extraction tool for archives, installers, logs, and backup files

Explanation of Why Each Software Is Needed

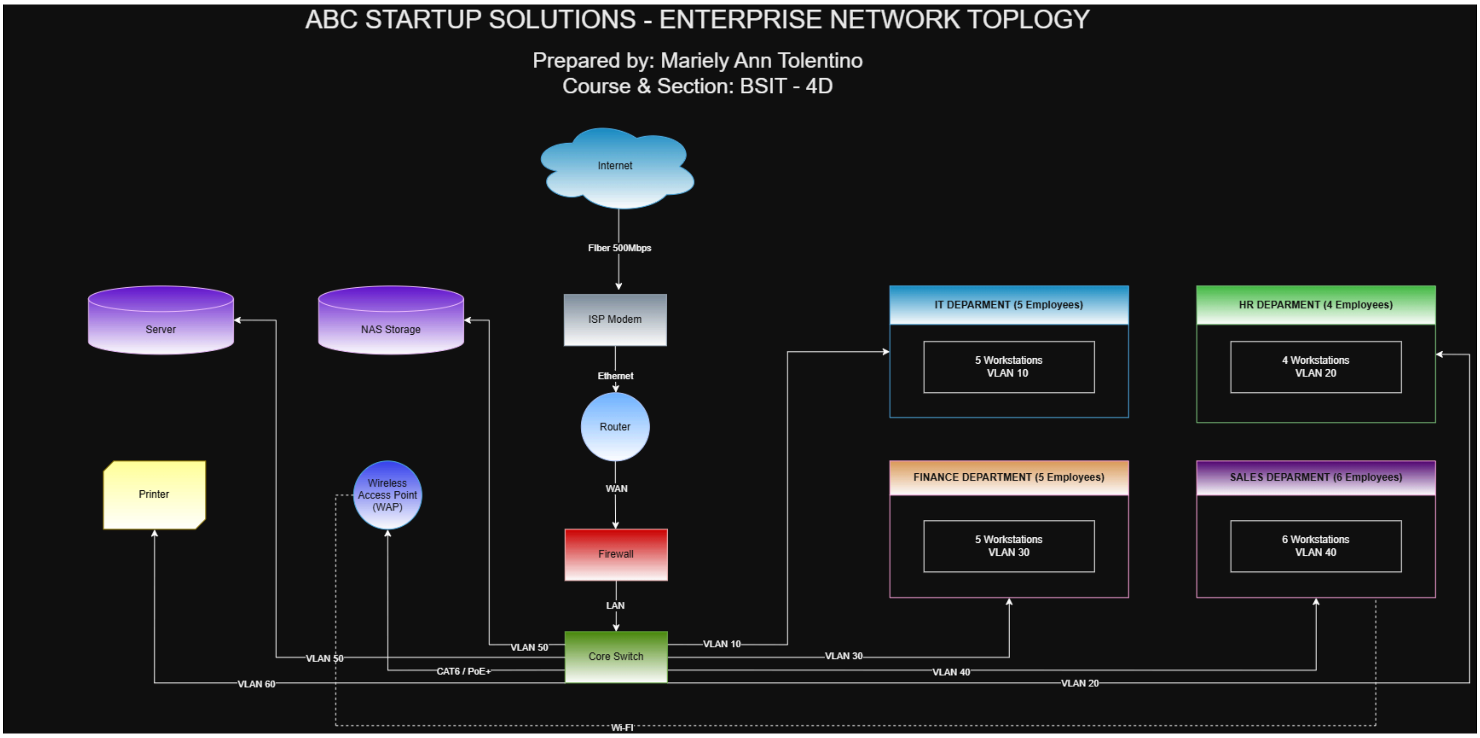
The listed software is needed to provide ABC Startup Solutions with a secure, productive, and efficient IT environment. Windows 11 Pro serves as the main operating system for company desktops and laptops because it offers business-ready features such as domain joining, BitLocker encryption, device management, and strong security support. Ubuntu Server is installed on the company server to centralize services such as file storage coordination, DNS, DHCP, backups, authentication support, monitoring, and internal tools. Microsoft Office supports daily productivity tasks across all departments by providing Word, Excel, PowerPoint, Outlook, and other office applications. VS Code is used by developers and IT staff as a lightweight code editor for programming, scripting, debugging, and configuration management. Git is required for version control so that multiple developers can track changes, collaborate, and maintain a complete history of project files. GitHub Desktop complements Git by providing a visual interface that makes basic repository tasks easier for users who are not comfortable with command-line tools. VirtualBox allows IT, developers, and QA staff to test operating systems, applications, and configurations inside virtual machines without affecting production computers. Google Chrome provides employees with a reliable browser for cloud applications, research, online tools, and web application testing, while also supporting enterprise security policies. Microsoft Defender protects Windows computers from viruses, malware, ransomware, and other threats through real-time endpoint protection. AnyDesk enables the IT department to provide secure remote support and troubleshoot user issues without needing to be physically present at each workstation. Finally, 7-Zip is used for compressing, extracting, packaging, and managing files, making it easier to handle archives, installers, project files, and backups.

PART 4: Enterprise Network Inventory

The table below lists the networking equipment required to build the company network from scratch.

ITEM	SPECIFICATION	QUANTITY	LOCATION	PURPOSE
ISP Modem	Fiber modem/ONT with gigabit Ethernet WAN port	1	IT Room / Rack	Terminates the ISP connection.
Router	Business-grade gigabit router with VLAN, VPN, and QoS support	1	Network Rack	Routes traffic between the office and the internet.
Firewall	Next-generation firewall (NGFW) or UTM with IPS and web filtering	1	Network Rack	Protects the network from external threats and unauthorized access.
Switch	48-port Gigabit PoE+ Managed Switch	1	Network Rack	Connects wired devices and provides PoE power to Access Points.
Access Point	Business Wi-Fi 6 / 6E Access Point, PoE-powered	2	Ceiling / Office Floor	Provides wireless coverage for laptops, phones, and guests.
Patch Panel	48-port CAT6 Patch Panel	1	Network Rack	Organizes and terminates structured cabling.
CAT6 Cables	CAT6 UTP Solid Copper Cable (305m box)	2 boxes	IT Storage / Cabling	Used for horizontal cabling to wall ports, desks, and APs.
RJ45 Connectors	CAT6 RJ45 Connectors (Pass-through or standard)	100 pcs	IT Storage	Used for terminating CAT6 cables.

PART 5: Enterprise Network Diagram



PART 6: System Administration Roles

Below is a detailed breakdown of four essential IT infrastructure roles, including their responsibilities, required skills, common tools, and industry-recognized certifications.

1. Helpdesk Technician

The Helpdesk Technician (or IT Support Specialist) is the frontline of the IT department, serving as the first point of contact for employees experiencing technical issues.

- **Responsibilities:**
  - Acting as the first line of support for hardware, software, and network issues.
  - Managing the IT ticketing system, logging incidents, and tracking resolutions.
  - Performing user onboarding and offboarding (creating accounts, assigning permissions, and setting up workstations).
  - Executing password resets, basic troubleshooting, and deploying standard software.
  - Escalating complex, unresolved issues to Tier 2 or Tier 3 administrators.
- **Skills:**
  - Excellent customer service and communication skills.
  - Strong foundational knowledge of Windows/macOS operating systems and basic networking.
  - Patience, empathy, and active listening.
  - Ability to translate technical jargon into simple terms for non-technical users.

- **Common Tools:**
  - Ticketing systems (Jira Service Management, Zendesk, ServiceNow, Spiceworks).
  - Remote support tools (AnyDesk, TeamViewer, BeyondTrust).
  - Identity management (Microsoft Active Directory / Entra ID).
- **Certifications:**
  - CompTIA A+
  - CompTIA IT Fundamentals (ITF+)
  - ITIL 4 Foundation (for IT service management)
  - Microsoft 365 Certified: Endpoint Administrator Associate

2. **Network Administrator**

The Network Administrator is responsible for the design, implementation, maintenance, and security of the organization's local and wide area networks (LAN/WAN/WLAN).

- **Responsibilities:**
  - Configuring and managing routers, switches, firewalls, and wireless access points.
  - Monitoring network performance, bandwidth utilization, and latency.
  - Designing and maintaining VLANs, subnets, and IP addressing schemes (DHCP/DNS).
  - Implementing network security policies, ACLs (Access Control Lists), and intrusion prevention systems.
  - Troubleshooting network outages and physical cabling issues.
- **Skills:**
  - Deep understanding of the OSI model, TCP/IP, and routing protocols (OSPF, BGP, EIGRP).
  - Knowledge of network security principles and firewall configuration.
  - Subnetting and VLAN trunking (802.1Q).
  - Packet analysis and network troubleshooting.
- **Common Tools:**
  - Network monitoring (SolarWinds, PRTG, Nagios, Zabbix).
  - Packet analyzers (Wireshark).
  - Terminal emulators (PuTTY, SecureCRT).
  - Network scanning (Nmap, Advanced IP Scanner).
- **Certifications:**
  - Cisco Certified Network Associate (CCNA)
  - CompTIA Network+
  - CompTIA Security+
  - Juniper Networks Certified Associate (JNCIA-Junos)

3. **Linux System Administrator**

The Linux System Administrator manages, configures, and maintains the organization's Linux-based servers, ensuring high availability, security, and performance for backend applications.

- **Responsibilities:**
    - Installing, configuring, and maintaining Linux servers (e.g., Ubuntu Server, RHEL, CentOS).
    - Managing user access, file permissions, and system security (SELinux, AppArmor, firewalld).
    - Writing scripts to automate routine tasks, backups, and system monitoring.
    - Managing web servers (Apache, Nginx), databases, and containerized environments.
    - Applying security patches, managing system updates, and monitoring server logs for anomalies.
- **Skills:**
    - Mastery of the Linux command-line interface (CLI) and shell environments.
    - Scripting proficiency (Bash, Python, Perl).
    - Understanding of system services (systemd), storage management (LVM), and virtualization (KVM).
    - Knowledge of configuration management and infrastructure automation.
- **Common Tools:**
    - Automation/Configuration (Ansible, Chef, Puppet).
    - Containerization (Docker, Podman).
    - Version control (Git).
    - Log management (ELK Stack: Elasticsearch, Logstash, Kibana).
- **Certifications:**
    - Red Hat Certified System Administrator (RHCSA)
    - CompTIA Linux+
    - Linux Foundation Certified System Administrator (LFCS)
    - Red Hat Certified Engineer (RHCE)

4. Cloud Administrator

The Cloud Administrator manages the organization's cloud infrastructure, ensuring that cloud resources are provisioned securely, optimized for cost, and highly available.

- **Responsibilities:**
    - Provisioning and managing cloud resources such as virtual machines, storage buckets, and managed databases (AWS, Azure, or GCP).
    - Managing Identity and Access Management (IAM) policies to ensure the principle of least privilege.
    - Monitoring cloud billing, optimizing resource usage, and reducing operational costs.
    - Implementing cloud backup strategies, disaster recovery, and high-availability architectures.
    - Automating cloud deployments using Infrastructure as Code (IaC).
- **Skills:**
    - Deep understanding of cloud service models (IaaS, PaaS, SaaS) and deployment models.
    - Proficiency in Infrastructure as Code (IaC) and cloud automation.
    - Cloud networking (VPCs, subnets, load balancers, gateways).
    - Cloud security, compliance, and cost management.
- **Common Tools:**

- Cloud Consoles (AWS Management Console, Azure Portal, GCP Console).
- Infrastructure as Code (Terraform, AWS CloudFormation, Azure Bicep).
- Container orchestration (Kubernetes, Amazon EKS, Azure AKS).
- Cloud monitoring (Amazon CloudWatch, Azure Monitor).

- **Certifications:**
  - AWS Certified SysOps Administrator Associate
  - Microsoft Certified: Azure Administrator Associate
  - Google Cloud Associate Cloud Engineer
  - HashiCorp Certified: Terraform Associate

How These Four Professionals Work Together

In a modern organization, these four IT professionals form a cohesive, multi-tiered ecosystem that ensures seamless business operations from the end-user device to the cloud backend. When an employee experiences a connectivity or application issue, the Helpdesk Technician acts as the first responder, resolving basic account or software problems and escalating complex tickets to the appropriate specialists. If the escalated issue is traced to network latency, a blocked port, or a failing switch, the Network Administrator steps in to troubleshoot the physical and logical network infrastructure to restore connectivity. Meanwhile, if a core internal application or database goes offline, the Linux System Administrator investigates the backend server logs, system resources, and services to restore functionality and ensure data integrity. Finally, as the company scales and integrates third-party SaaS platforms or migrates heavy workloads off-premises, the Cloud Administrator ensures that these cloud-hosted resources communicate securely with the on-premises network, managing cloud infrastructure, IAM permissions, and cost optimization. Together, they create a unified IT environment where user experience, network stability, server reliability, and cloud scalability are perfectly balanced.

PART 7: Infrastructure Recommendations

1. Internet Provider

Recommendation:

Implement a Dual-WAN (Redundant) Internet Setup.

- **Primary Connection:** A dedicated Enterprise Fiber line (e.g., PLDT Enterprise, Globe Business, or Converge Business) with a minimum speed of 300 to 500 Mbps and a static IP address.
- **Secondary/Backup Connection:** A 5G/LTE Business Router (e.g., Smart or Globe 5G) or a secondary fiber line from a different ISP to serve as an automatic failover.

Logical Reasons:

- **Zero Downtime for Developers:** As a software development company, our IT team relies heavily on continuous internet access to push code to GitHub, pull Docker images, and deploy cloud applications. An internet outage halts production entirely.
- **Client Communication:** The Sales department requires uninterrupted connectivity for video conferences and CRM access during client pitches. A failover connection ensures business continuity if the primary ISP experiences regional outages in the Nuvali/Santa Rosa area.

2. Server Specifications

Recommendation:

Deploy an Enterprise-Grade Tower or Rackmount Server (e.g., Dell PowerEdge T350 or HPE ProLiant MicroServer) with the following baseline specifications:

- **Processor:** Intel Xeon E-Series or AMD EPYC (Minimum 8 Cores).
- **Memory:** 64GB ECC (Error-Correcting Code) RAM.
- **Storage (OS):** 2x 1TB NVMe SSDs in RAID 1 (Mirroring).
- **Storage (Data/VMs):** 4x 4TB Enterprise HDDs in RAID 5 or RAID 10.

Logical Reasons:

- **ECC RAM:** Essential for servers running Ubuntu Server and hosting databases or virtual machines. It detects and fixes internal data corruption, preventing system crashes and silent data degradation.
- **RAID Configuration:** RAID 1 ensures the server remains operational even if an OS drive fails. RAID 5/10 for data ensures that we do not lose local repositories, backups, or shared files if a hard drive physically fails.
- **Virtualization Ready:** High core counts and 64GB RAM allow the IT team to run VirtualBox or KVM virtual machines locally for testing environments without degrading server performance.

3. Backup Strategy

Recommendation:

Strictly implement the industry-standard 3-2-1 Backup Strategy, managed via automated backup software (e.g., Veeam Backup & Replication or UrBackup).

- **3 Copies of Data:** 1 Production copy, 2 Backup copies.
- **2 Different Media:** Local NAS (Network Attached Storage) and External USB Hard Drives.
- **1 Offsite/Cloud:** Encrypted cloud backup (e.g., AWS S3, Backblaze B2, or Microsoft OneDrive for Business).

Logical Reasons:

- **Ransomware Protection:** If the primary network is infected with ransomware, local backups on the NAS might also be encrypted. Having an offline external drive (rotated weekly) and an immutable cloud backup guarantees we can restore our data without paying a ransom.
- **Disaster Recovery:** In the event of a physical disaster (e.g., fire, flood, or severe typhoon in Laguna), the offsite cloud backup ensures the company's intellectual property and financial records survive and can be restored remotely.

4. Security Recommendations

Recommendation:

Adopt a Defense-in-Depth approach focusing on network segmentation and physical security.

- **VLAN Segmentation:** Separate the network into VLANs (e.g., VLAN 10: Management, VLAN 20: Servers, VLAN 30: HR/Finance, VLAN 40: IT/Dev, VLAN 50: Sales, VLAN 60: Guest Wi-Fi).
- **BitLocker Encryption:** Enforce full-disk encryption on all Windows 11 Pro desktops and laptops.
- **Physical Security:** House the router, firewall, switch, and server in a locked, ventilated network rack in a restricted-access IT room.

Logical Reasons:

- **Containment of Breaches:** If a Sales employee clicks a malicious link on the Guest or Sales VLAN, VLAN segmentation prevents the malware from laterally moving into the Finance or Server VLANs where sensitive payroll and source code data reside.
- **Data at Rest Protection:** Sales laptops frequently leave the office for client meetings. If a laptop is lost or stolen, BitLocker ensures the thief cannot extract company data or client proposals from the hard drive.

5. Antivirus

Recommendation:

Utilize Microsoft Defender for Endpoint (Business/Enterprise Tier) as our primary antivirus and Endpoint Detection and Response (EDR) solution, managed centrally via the Microsoft 365 Defender portal.

Logical Reasons:

- **Cost-Effective & Native:** Since we are already deploying Windows 11 Pro and Microsoft 365, Defender is natively integrated, requiring no third-party agent installations that could slow down developer machines.
- **Centralized Visibility:** The Junior System Administrator can view the health, threat alerts, and update status of all 20 endpoints from a single web dashboard. It provides enterprise-grade protection against zero-day threats, ransomware, and phishing attacks without additional licensing costs.

6. Password Policy

Recommendation:

Align with modern NIST (National Institute of Standards and Technology) Guidelines:

- **Length over Complexity:** Require a minimum of 12 characters (encouraging passphrases like **Correct-Horse-Battery-Staple!**) rather than forcing complex, short passwords.
- **No Mandatory Expiration:** Do not force password changes every 90 days unless a breach is suspected (which prevents users from writing passwords on sticky notes).
- **Mandatory MFA:** Enforce Multi-Factor Authentication (MFA) via Microsoft Authenticator for all email, cloud, and remote access logins.
- **Enterprise Password Manager:** Deploy a corporate password manager (e.g., Bitwarden or 1Password).

Logical Reasons:

- **Defeating Automated Attacks:** MFA stops 99.9% of automated credential-stuffing and phishing attacks. Even if a hacker guesses a password, they cannot access the account without the user's mobile device.
- **Eliminating Password Reuse:** A password manager allows employees to generate and store unique, 20-character passwords for GitHub, AWS, banking, and email, eliminating the dangerous habit of using the same password across multiple platforms.

7. Expansion Plan

Recommendation:

Design the current infrastructure to support a Hybrid-Cloud Transition and 50-User Scalability over the next 3 years.

- **Phase 1 (Year 1):** Stabilize the on-premises network, enforce security policies, and establish reliable local backups.
- **Phase 2 (Year 2):** Transition heavy file storage to Microsoft SharePoint/OneDrive to support remote/hybrid work models. Introduce a Zero Trust Network Access (ZTNA) framework.
- **Phase 3 (Year 3+):** Upgrade the core switch to a multi-switch stacked environment and migrate local CI/CD pipelines and internal Git repositories to cloud-hosted equivalents (e.g., GitHub Enterprise or AWS CodePipeline) if the developer team expands significantly.

Logical Reasons:

- **Future-Proofing Investment:** By currently purchasing a 48-port PoE+ switch and running CAT6 cabling, the company can easily double its workforce to 40 employees without needing to rip and replace the physical network infrastructure.
- **Supporting Remote Work:** As the company grows, top-tier software developers may prefer remote or hybrid work arrangements. Transitioning to cloud-based identity (Entra ID) and file storage now ensures that future off-site employees will have the same secure, fast access to resources as those sitting in the Laguna office.

PART 8: Personal Reflection

Designing a complete IT infrastructure from scratch for ABC Startup Solutions has been one of the most eye-opening experiences in my journey toward becoming a System Administrator. Through this project, I learned that enterprise IT is not merely about purchasing computers and connecting them to the internet; it is about strategically aligning technology with the specific operational needs of every department. I gained practical knowledge in selecting enterprise hardware, managing software licensing, structuring VLANs, and establishing robust security and backup policies. Most importantly, I learned how each individual component from the ISP modem to the end-user's dual monitors must work together seamlessly to create a secure, scalable, and efficient working environment.

The most challenging task for me was designing the Enterprise Network Diagram and formulating the Infrastructure Recommendations. Translating theoretical networking concepts into a logical, visual topology required careful consideration of traffic flow, security boundaries, and physical cabling limits. Deciding how to segment the IT, HR, Finance, and Sales departments into different VLANs while ensuring they could still securely access the central server forced me to think critically about both connectivity and data isolation. Additionally, justifying the hardware specifications and backup strategies required me to balance ideal enterprise standards with the realistic financial constraints and growth projections of a 20-person startup.

This project deeply reinforced the fact that planning is absolutely critical before deployment. In a real-world scenario, deploying hardware without a comprehensive blueprint often leads to costly rework, network bottlenecks, and severe security vulnerabilities. By planning ahead, we can accurately forecast budgets, ensure IP address scalability, and establish disaster recovery protocols before a crisis actually occurs. A well-documented plan acts as a roadmap that minimizes downtime during installation and provides a reliable baseline for future troubleshooting. I realized that a reactive approach to IT is exhausting and expensive, whereas a proactive, planned approach builds organizational trust and operational stability.

Ultimately, this project will help me become a significantly better System Administrator by shifting my mindset from that of a reactive troubleshooter to a strategic infrastructure architect. I now understand that technical skills alone are not enough; a great administrator must also understand business workflows, risk management, and user experience. The ability to document, justify, and design systems from the ground up will allow me to approach future enterprise projects with confidence, ensuring that the technology I manage not only keeps the company running today but is fully prepared to support its long-term growth tomorrow.